

Answer Key: Lesson 01 Square and Cube Roots

Question identifiers match the student documents exactly. **Notation:** $\sqrt{n}$ is the non-negative square root; $\approx$ marks an estimate to 1 decimal place (1 d.p.). All 1 d.p. estimates have been checked against the true value (shown in brackets) so that the “closer” rule gives the correct rounding. *Acceptable alternative for any estimate:* test the halfway value, e.g. $7.05^2 = 49.7025 < 50$, so $\sqrt{50}$ rounds up to 7.1.

1. Preparation (Lesson_01 Preparation)

P1–P9

P1. (a) $6^2 = 36$ (b) $9^2 = 81$ (c) $12^2 = 144$ (d) $2^3 = 8$ (e) $4^3 = 64$

P2. $5^3 = 5 \times 5 \times 5 = 25 \times 5 = 125$

P3. 25 (5^2), 36 (6^2), 81 (9^2). Not 20, 50, 90 (between squares, e.g. $49 < 50 < 64$).

P4. Both: $64 = 8 \times 8 = 8^2$ and $64 = 4 \times 4 \times 4 = 4^3$.

P5. n^2 : 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144. n^3 ($n = 1$ to 5): 1, 8, 27, 64, 125.

P6. $(-5)^2 = (-5) \times (-5) = 25$.

P7. $(-3)^3 = (-3) \times (-3) \times (-3) = 9 \times (-3) = -27$.

P8. (a) $8 \times 8 = 64 \text{ cm}^2$ (b) $3 \times 3 \times 3 = 27 \text{ cm}^3$.

P9. (a) 6 cm, because $6 \times 6 = 36$. (b) Not a whole number: $6^2 = 36 < 40 < 49 = 7^2$, so the side is between 6 and 7 cm (nearer 6; $\approx 6.3 \text{ cm}$).

Two-minute plan for common preparation errors (use on slide 4, 0:05–0:08)

$3^3 = 9$ or $5^3 = 15$

Write in full on the board: $3^3 = 3 \times 3 \times 3 = 27$. “The small number tells you how many copies, not what to multiply by.”

$(-3)^3 = 27$;
 $(-5)^2 = -25$

Pair the signs: $(-)(-) = (+)$; a third $(-)$ makes it negative again.
Squares of negatives are positive.

P9(b): “20” or “no answer”

“Side $\times$ side = 40. Try 6×6 and 7×7 .” Leads straight into N1.

2. Starter (Class Worksheet p.1)

S1–S6

S1. 49 **S2.** 121 **S3.** 27 **S4.** 1000 **S5.** 16 **S6.** 64 (8^2 and 4^3). 16 and 100 are squares only; 27 is a cube only.

3. Mini-whiteboards

Rounds 1 and 2

Round 1: Q1 $\sqrt{9} = 3$; Q2 $\sqrt[3]{8} = 2$; Q3 $\sqrt{64} = 8$; Q4 $\sqrt[3]{-125} = -5$ ($(-5)^3 = -125$); Q5 False: $\sqrt[3]{27} = 3$ ($9^3 = 729$). Re-check: $\sqrt{36} = 6$, $\sqrt[3]{1000} = 10$.

Round 2: Q1 3 and 4 ($9 < 10 < 16$); Q2 6 and 7 ($36 < 45 < 49$); Q3 2 and 3 ($8 < 20 < 27$); Q4 $\sqrt{30} \approx 5.5$ ($5.4^2 = 29.16$, $5.5^2 = 30.25$; 30 is 0.25 from 30.25, 0.84 from 29.16; true 5.477); Q5 False: $(-2)^2 = 4$, so $\sqrt{-4}$ is not a real number. Reteach: $\sqrt{70} \approx 8.4$ ($8.3^2 = 68.89$, $8.4^2 = 70.56$; true 8.367). Re-check: $\sqrt{12}$ is between 3 and 4.

4. Notice/Why, I DO and WE DO

N1, Ex 1–3, W1–W3

N1. (a) $\sqrt{16} = 4$, $\sqrt{25} = 5$. (b) Between 4 cm and 5 cm, because $16 < 20 < 25$. Slightly closer to 4, because 20 is 4 away from 16 but 5 away from 25 (true side ≈ 4.47 cm). Accept “about halfway, a little nearer 4”.

Example 1: (a) 9 (b) 5 (c) -2 (d) no real answer. **Example 2:** $\sqrt{50} \approx 7.1$ (7.071). **Example 3:** $\sqrt[3]{30} \approx 3.1$ (3.107). Full working is printed on the worksheet.

W1. (a) $\sqrt{144} = 12$ because $12^2 = 144$. (b) $\sqrt[3]{64} = 4$ because $4^3 = 64$. (c) $\sqrt[3]{-27} = -3$ because $(-3)^3 = -27$.

W2. Trap $16 < 20 < 25$; Roots $4 < \sqrt{20} < 5$; Closer to: 16, but only just (4 vs 5 away), so test. Test $4.4^2 = 19.36$, $4.5^2 = 20.25$. Decide: $20.25 - 20 = 0.25 < 20 - 19.36 = 0.64$, so $\sqrt{20} \approx 4.5$ (4.472). *Note:* the whole-number “closer” step points slightly towards 4, yet the answer rounds to 4.5. This is why the Test step is essential. Accept “Closer to: 4” or “16” in the blank.

W3. $64 < 100 < 125 \Rightarrow 4 < \sqrt[3]{100} < 5$. 100 is 36 from 64 and 25 from 125, so a little nearer 5. $4.6^3 = 97.336$, $4.7^3 = 103.823$. $100 - 97.336 = 2.664 < 3.823$, so $\sqrt[3]{100} \approx 4.6$ (4.642).

5. Class Worksheet: YOU DO

A1–D3

A. Foundation. A1 5 A2 10 A3 11 A4 2 A5 10 A6 -4 ($(-4)^3 = -64$) A7 13

A8. $4 < \sqrt{18} < 5$ because $16 < 18 < 25$.

B. Core. B1 15 B2 6 B3 -5

B4. Any real number squared is positive or zero: $4^2 = 16$ and $(-4)^2 = 16$. No real number squared gives -16 , so $\sqrt{-16}$ is not a real number.

B5. $27 < 40 < 64$, so $3 < \sqrt[3]{40} < 4$.

B6. $9 < 10 < 16 \Rightarrow 3 < \sqrt{10} < 4$, near 3. $3.1^2 = 9.61$, $3.2^2 = 10.24$; 10 is 0.24 from 10.24, 0.39 from 9.61, so $\sqrt{10} \approx 3.2$ (3.162).

B7. $64 < 75 < 81 \Rightarrow 8 < \sqrt{75} < 9$, nearer 9. $8.6^2 = 73.96$, $8.7^2 = 75.69$; 0.69 vs 1.04, so $\sqrt{75} \approx 8.7$ (8.660).

B8. $8 < 20 < 27 \Rightarrow 2 < \sqrt[3]{20} < 3$, nearer 3. $2.7^3 = 19.683$, $2.8^3 = 21.952$; 0.317 vs 1.952, so $\sqrt[3]{20} \approx 2.7$ (2.714).

C. Apply and reason. C1. $\sqrt{196} = 14$ m ($14^2 = 196$). C2. $\sqrt[3]{343} = 7$ cm ($7^3 = 343$).

C3. $\sqrt{40}$: $36 < 40 < 49$, $6.3^2 = 39.69$, $6.4^2 = 40.96$, so $\sqrt{40} \approx 6.3$. $\sqrt[3]{200}$: $125 < 200 < 216$, so between 5 and 6, near 6; $5.8^3 = 195.112$, $5.9^3 = 205.379$, so ≈ 5.8 . **Order:** $\sqrt[3]{200}$, 6, $\sqrt{40}$, 6.5. (Marks at about 5.8, 6, 6.3, 6.5 on the number line.)

C4. $49 < 60 < 64 \Rightarrow 7 < \sqrt{60} < 8$. $7.7^2 = 59.29$, $7.8^2 = 60.84$; 0.71 vs 0.84, so side ≈ 7.7 cm (7.746). Check: $7.7^2 = 59.29 \approx 60$ ✓. *Alternative:* $7.75^2 = 60.0625 > 60$, so rounds down to 7.7.

C5. Cube edge = $\sqrt[3]{64} = 4$ cm; square side = $\sqrt{64} = 8$ cm. The square has the longer side (8 cm $>$ 4 cm).

D. Challenge. D1. (a) $\sqrt{\frac{9}{16}} = \frac{\sqrt{9}}{\sqrt{16}} = \frac{3}{4}$ (check $\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$). (b) $\sqrt[3]{\frac{1}{8}} = \frac{1}{2}$ (check $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$).

D2. $\sqrt{0.25} = 0.5$ (since $0.5^2 = 0.25$), and $0.5 > 0.25$: bigger. $\sqrt{1} = 1$: the same. Maya is wrong: for numbers between 0 and 1 the square root is larger; it is only smaller for numbers greater than 1.

D3. Need $9.5^2 < n < 10^2$, i.e. $90.25 < n < 100$. Any of $n = 91, 92, \dots, 99$: **9** whole numbers.

6. Error Analysis: Spot the Mistake**E1–E3**

E1. First error: “ $36 \div 2$ ”. Sam halved instead of finding a number that times itself gives 36. Correct: $\sqrt{36} = 6$ since $6^2 = 36$ (check: $18^2 = 324$).

E2. First error: $\sqrt{9 + 16} = \sqrt{9} + \sqrt{16}$. A root cannot be split over addition. Correct: $\sqrt{9 + 16} = \sqrt{25} = 5$.

E3. First error: “ $\sqrt{40}$ is halfway”. 40 is 4 away from 36 and 9 away from 49, so it is nearer 36 and $\sqrt{40}$ is nearer 6. Test: $6.3^2 = 39.69$, $6.4^2 = 40.96$; $\sqrt{40} \approx 6.3$ (6.325).

7. Exit Ticket**X1–X4**

X1. (a) $\sqrt{49} = 7$ (b) $\sqrt[3]{27} = 3$ (SC1)

X2. $\sqrt[3]{-1000} = -10$. Check: $(-10) \times (-10) \times (-10) = 100 \times (-10) = -1000$ ✓ (SC2)

X3. (a) $81 < 85 < 100$, so $9 < \sqrt{85} < 10$. (b) Nearer 81. $9.2^2 = 84.64$, $9.3^2 = 86.49$; 0.36 vs 1.49, so $\sqrt{85} \approx 9.2$ (9.220). (SC3, SC4)

X4. Edge = $\sqrt[3]{90}$. $64 < 90 < 125$ (4^3 and 5^3), so the edge is between 4 cm and 5 cm. (Using squares, “9 and 10”, is the key error to look for.) (transfer)

8. After-Lesson Practice**R1, H1–H16**

R1. (a) $10^3 = 1000$ (b) $(-2)^4 = (-2)(-2)(-2)(-2) = 16$.

Easy. H1. (a) 6 (b) 9 (c) 12 **H2.** (a) 3 (b) 10 (c) -2 **H3.** $\sqrt{169} = 13$ because $13^2 = 169$.

H4. (a) $3 < \sqrt{12} < 4$ ($9 < 12 < 16$) (b) $8 < \sqrt{70} < 9$ ($64 < 70 < 81$) (c) $2 < \sqrt[3]{10} < 3$ ($8 < 10 < 27$).

H5. $\sqrt{64} = 8$ cm. **H6.** $\sqrt[3]{216} = 6$ cm ($6^3 = 216$).

Medium. H7. $36 < 42 < 49 \Rightarrow 6 < \sqrt{42} < 7$. $6.4^2 = 40.96$, $6.5^2 = 42.25$; 0.25 vs 1.04, so $\sqrt{42} \approx 6.5$ (6.481).

H8. $27 < 60 < 64 \Rightarrow 3 < \sqrt[3]{60} < 4$, much nearer 4. $3.9^3 = 59.319$, $4^3 = 64$; 0.681 vs 4, so $\sqrt[3]{60} \approx 3.9$ (3.915).

H9. $26 > 25$, so $\sqrt{26} > \sqrt{25} = 5$. $100 < 125$, so $\sqrt[3]{100} < \sqrt[3]{125} = 5$. Hence $\sqrt{26} > 5 > \sqrt[3]{100}$: $\sqrt{26}$ is larger.

H10. $(-4)^3 = -64$, so $\sqrt[3]{-64} = -4$ is real. But any real number squared is ≥ 0 (e.g. $8^2 = 64$, $(-8)^2 = 64$), so no real number squares to -64; $\sqrt{-64}$ is not real.

H11. $\sqrt[3]{9}$: $8 < 9 < 27$, just over 2 ($2.1^3 = 9.261 > 9$, so below 2.1). $\sqrt{5}$: $4 < 5 < 9$, just over 2 ($2.2^2 = 4.84 < 5$, so above 2.2). $\sqrt[3]{27} = 3$. $\sqrt{15}$: $9 < 15 < 16$, just under 4 (≈ 3.9). **Order:** $\sqrt[3]{9}$ (≈ 2.1), $\sqrt{5}$ (≈ 2.2), $\sqrt[3]{27} = 3$, 3.5, $\sqrt{15}$ (≈ 3.9).

H12. (a) 400 (20^2) (b) -27 ($(-3)^3$) (c) 0.25 (0.5^2).

Hard. H13. Side = $\sqrt{30}$. $5.2^2 = 27.04 < 30$, so $\sqrt{30} > 5.2$ (in fact ≈ 5.5 m, since $5.4^2 = 29.16$, $5.5^2 = 30.25$). The rug is wider than the hallway, so it will **not** fit flat across.

H14. (a) First mistake: assuming $\sqrt[3]{70}$ is halfway because 70 is between 64 and 125. 70 is only 6 from 64 but 55 from 125. (b) $4.1^3 = 68.921$, $4.2^3 = 74.088$; 70 is 1.079 from 68.921 and 4.088 from 74.088, so $\sqrt[3]{70} \approx 4.1$ (4.121). Check: $4.1^3 \approx 69 \approx 70$ ✓.

H15. $6 < \sqrt{n} < 7 \iff 36 < n < 49$. So $n = 37, 38, \dots, 48$: **12** numbers. All found because every whole number strictly between the neighbouring perfect squares 36 and 49 is included ($48 - 37 + 1 = 12$). (If a student includes 36 and 49 and reads “between” inclusively, accept 14 with that reasoning stated.)

H16. Volume = $4 \times 4 \times 8 = 128 \text{ cm}^3$. $125 < 128 < 216 \Rightarrow 5 < \sqrt[3]{128} < 6$, very close to 5. $5.0^3 = 125$, $5.1^3 = 132.651$; 3 vs 4.651, so edge $\approx 5.0 \text{ cm}$ (5.040).

9. Self-Test

ST1–ST3

ST1. $\sqrt{121} = 11$ **ST2.** $\sqrt[3]{-216} = -6$ ($(-6)^3 = -216$) **ST3.** $49 < 55 < 64$; $7.4^2 = 54.76$, $7.5^2 = 56.25$; 0.24 vs 1.25, so $\sqrt{55} \approx 7.4$ (7.416).